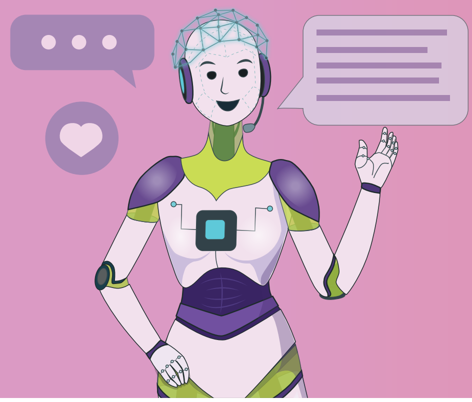
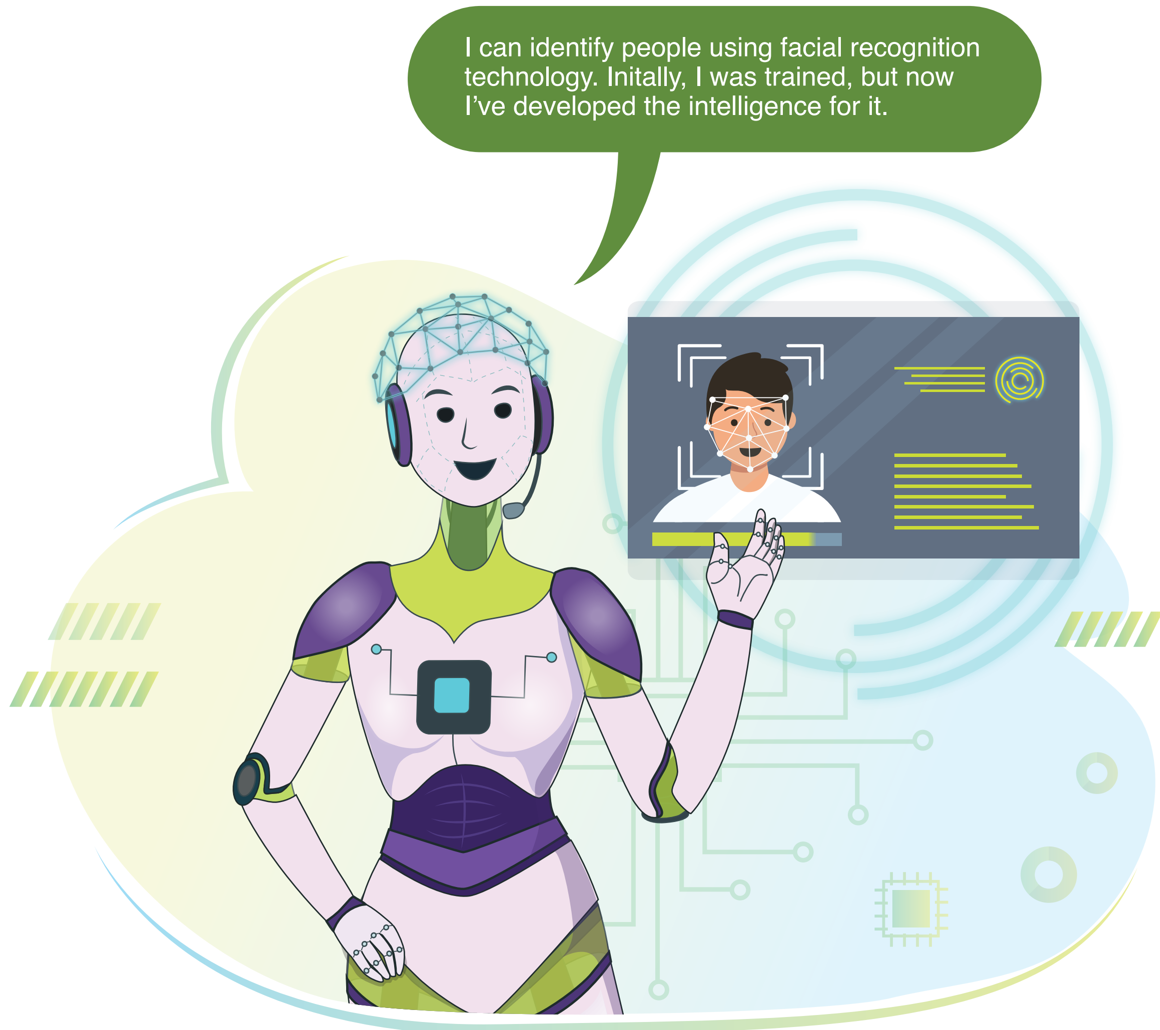


Introduction to AI

16



Have you ever used the voice assistant feature on a smartphone? What have you asked the assistant to do for you? A voice assistant constantly learns from our commands. Based on the conversations we have, it stores our preferences for future use.

Out of the hundreds of conversations we have with it, a voice assistant cannot store all the data we enter.

It uses a new programming method which is based on "training" computers to learn. The computer learns based on inputs given by the user. Since it evolves and learns step by step after each iteration, this programming approach is called **"Machine Learning"**.

Machine learning is a branch of Artificial Intelligence (AI).

Intelligence can be categorised into two:

1. **Natural intelligence** is demonstrated by sentient beings like humans and animals who use their senses to see and feel.

2. **Artificial intelligence** is demonstrated by a machine or computer programmed with an evolved consciousness. It exhibits emotional and logical intelligence similar to the human mind.

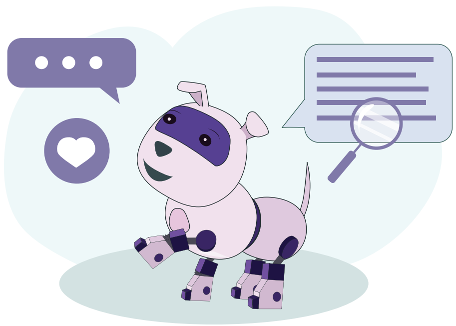
Relevance and applications of AI

Artificial intelligence has a myriad of applications:

1. **AI in marketing:** Using AI, marketers can deliver highly targeted and personalised ads with the help of behavioural analysis, pattern recognition, etc. AI provides users with real-time personalisation based on their behaviour. It is used to edit and optimize marketing campaigns to fit a local market's needs. For example, if we search for a product on the internet, the AI will keep track of our searches and then show us ads for similar products.

2. **AI in automobiles:** AI is being used to build self-driving vehicles. It works with a vehicle's camera, its radar, cloud services, GPS, and control signals to operate the vehicle. AI also enables a vehicle to self-diagnose itself.

3. **AI in social media:** AI models are used to optimise our social media feeds according to our searches, likes and the accounts we follow. They are used to curate ads shown to users on social media platforms.



4. **AI in e-commerce:** AI technology is used to create recommendation engines through which businesses can better engage with their customers. They can track buying history, wish lists and products viewed. This enables them to recommend products which customers are likely to buy as well as provide combo offers.

5. **AI in robotics:** Robots are powered by AI using real-time updates to sense obstacles and pre-plan their journey. They are trained to perform household work and communicate with users.

6. **AI in gaming:** AI is used to predict human behaviour. This improves game design and testing. AI bots can combat with players in multiplayer arcade games and role player games.

Complex algorithms and layers are used to design these systems.

When it comes to operating AI, not everyone is technically equipped to create programs from scratch and simulate results. Instead, AI engines are developed to provide a more general-purpose solution which can be understood by everyone.

AI engines: An AI engine is a tool that helps build an artificially intelligent system. It is a prebuilt program which takes in a generic input and provides a solution. For example, an image recognition-based engine, a speech-based engine, a chatbot, etc.

AI engines are developed and distributed for free by companies: For example, IBM's Watson, Google's TensorFlow, Amazon Lex.

Individuals and companies can learn to operate these AI engines and build applications with them.

Machine Learning for kids : ML for kids is a free tool which introduces children to basic concepts of machine learning through interactive projects by letting them train models to recognize text, numbers, images, or sounds in an easy-to-follow environment. It's added to coding platforms like Scratch and App Inventor.

Exercise

Fill in the blanks

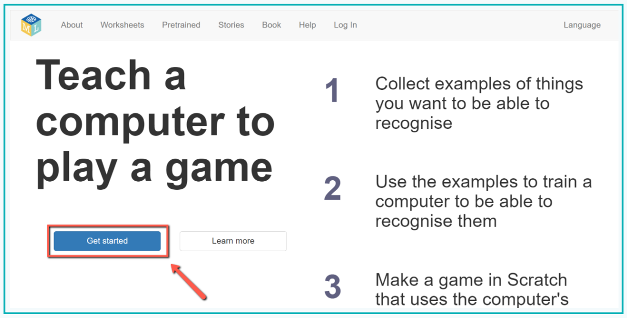
1. __________________ is a new programming method of "training" computers to learn.
2. An__________________ is a module that helps us build an artificially intelligent system.
3. ___________________ is a free tool which introduces children to basic concepts of machine learning through interactive projects.

Activity

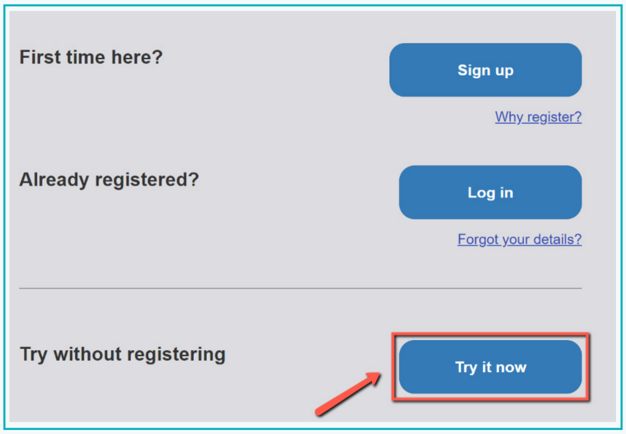
Build an AI application to identify things such as Car or Cup.

1 Visit https://machinelearningforkids.co.uk/

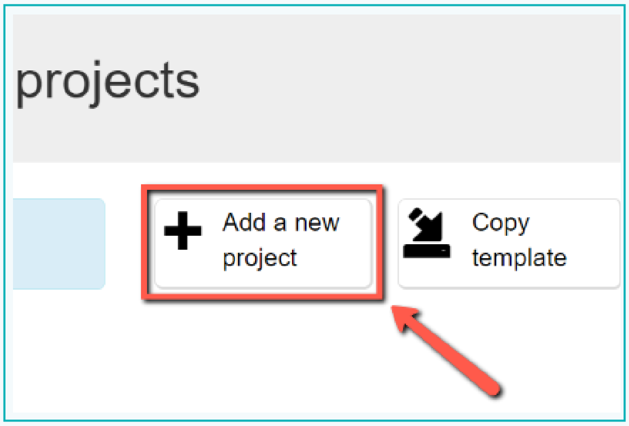
- Click on: “Get Started”.



- There will be three options, among them select ‘Try without registering’ by clicking on “Try it now”.

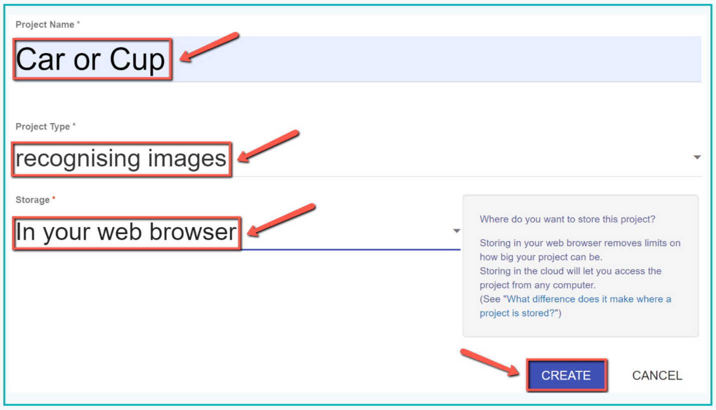


2 Click the “+ Add new project” button.

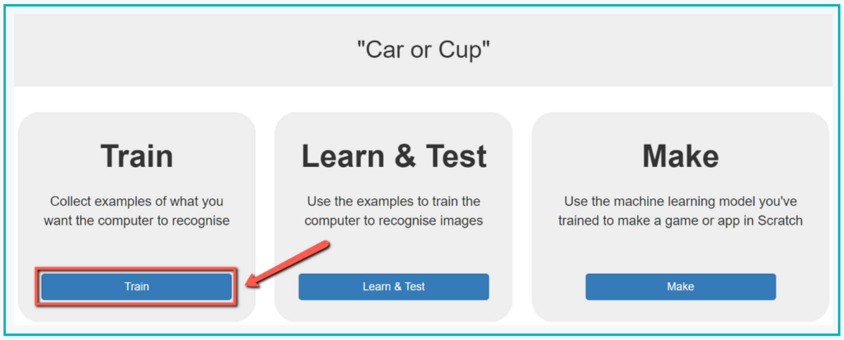


Activity

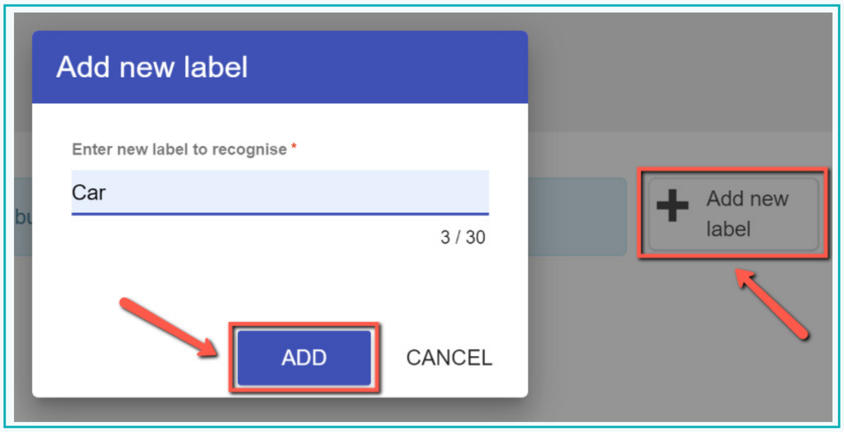
3 Name your project “Car or Cup” and set project type to “recognising images”. Then click the “Create” button.



4 You should now see “Car or Cup” in the list of your projects. Click on it & select “Train” button to start collecting examples.

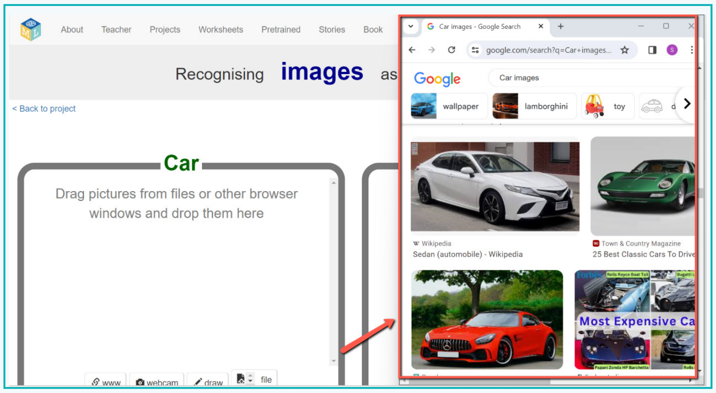


5 Click on “+ Add new label” and call it “Car”. Do that again and create a second bucket called “Cup”.



Activity

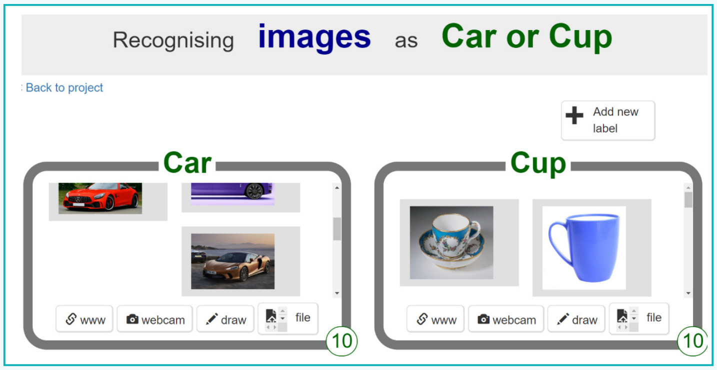
6. Open another web browser window & arrange the web browser windows so that they are side by side.



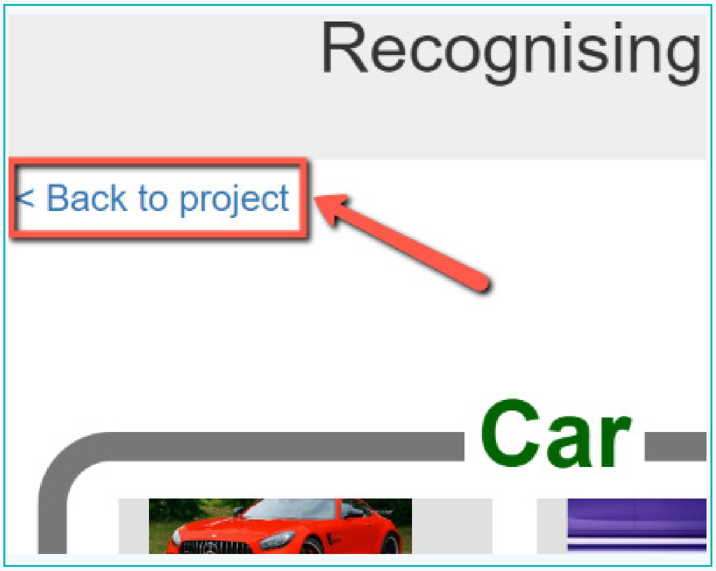
a. In the new browser window, search for pictures of cars and cups.

b. Drag pictures that are good examples of cars and cups into respective buckets.

c. Repeat until you get 10 examples of car photos and cup photos.



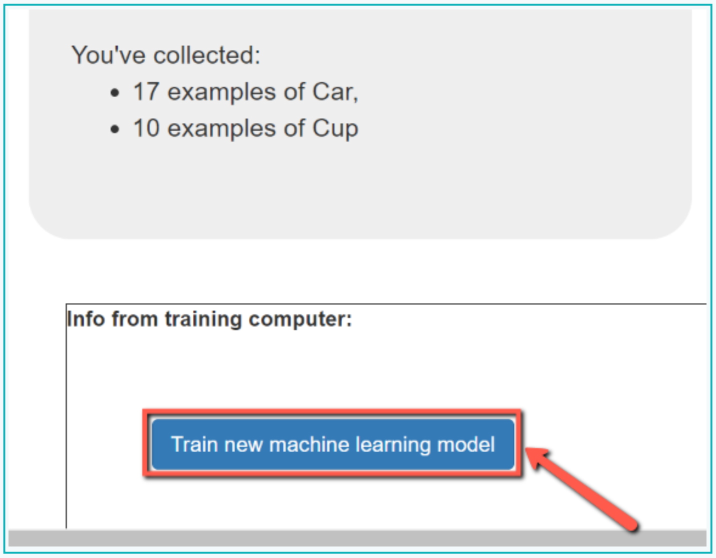
7. Click the “< Back to project” link.



Activity

8 Click the “Learn & Test” button & click the “Train new machine learning model” button.

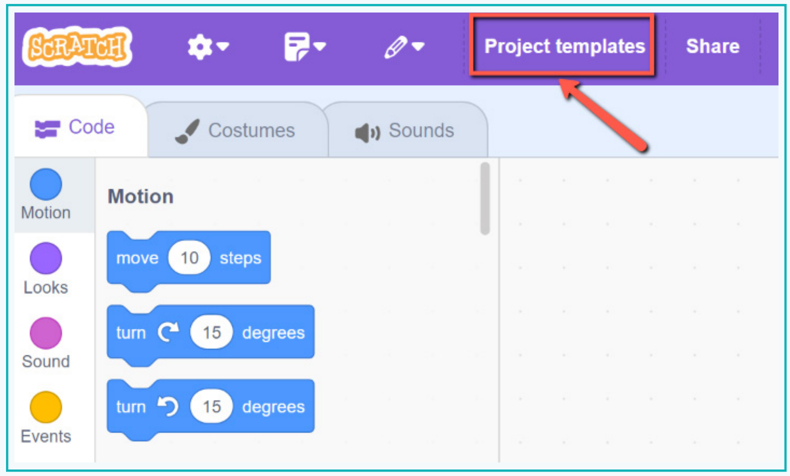
- Wait for the training to complete.
- This might take a few minutes.



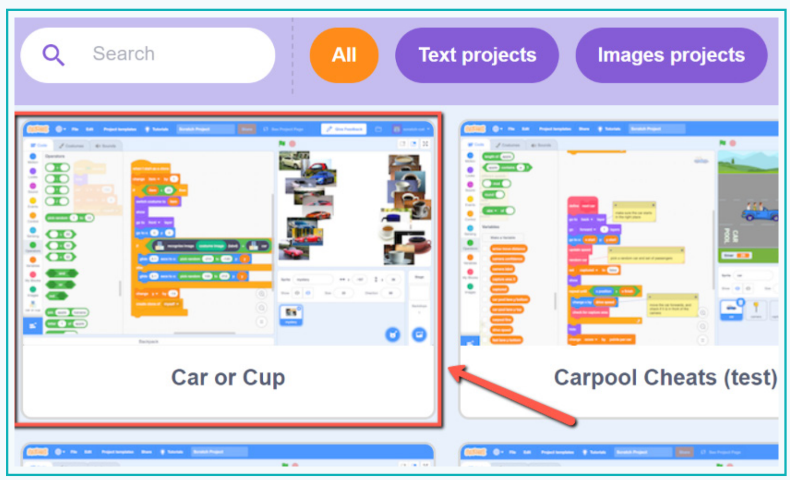
9 Click the “< Back to project” link & click the “Make” button, and then the “Scratch 3” button.

Activity

10. Click on “Project templates”.



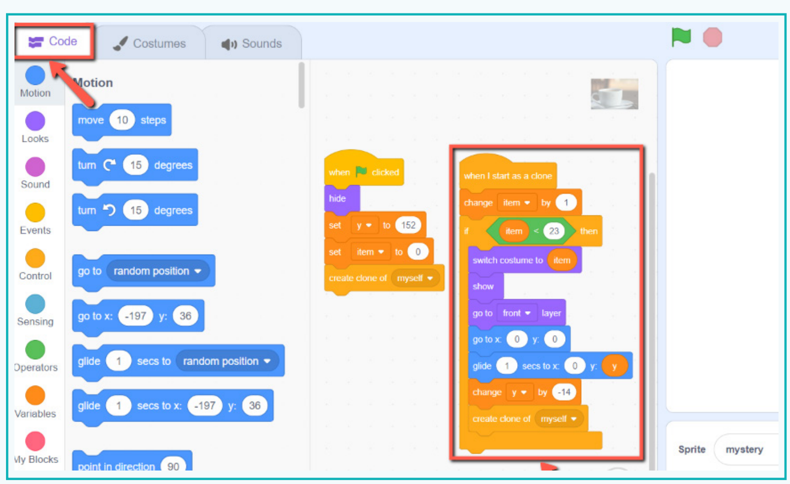
11. Then select Car or Cup template.



12. Click on the “mystery” sprite, then the “Code” tab.

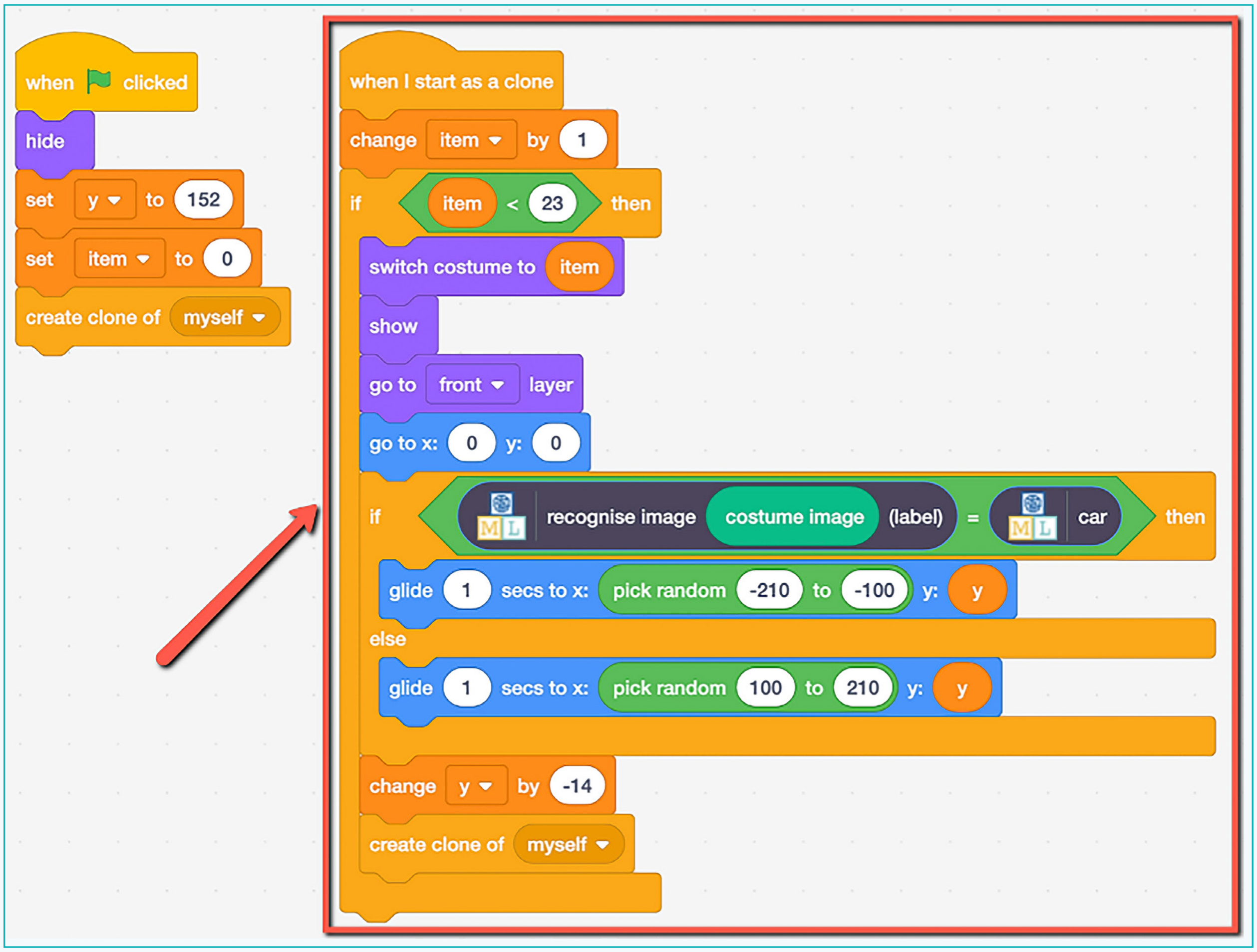
    a. Change the script to use your machine learning model.

    b. Start from the script that is already there.



Activity

c. Change it to look like this.



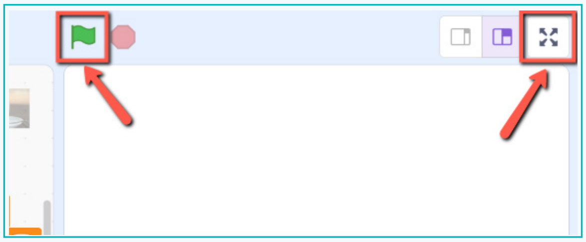
d. You can easily find respective blocks on the left side of the screen.

e. You need to drag and drop respective blocks to arrange code in a mentioned manner.

f. Right click on the block to delete or duplicate the block.

g. Please refer to the shape of the blocks in order to arrange it in respective alignment.

13 Click the full screen icon, and then click the green flag.



Activity

Watch your script use your model to sort the photos into two piles.

Note:
If your trained system makes mistakes, you'll need to go back to step 7, and collect more examples. But make sure you repeat step 10 to train a new model.

Tips:

- The more examples you give in training, the better the computer should get at recognising whether a photo is a cup or car.
- Try and come up with roughly the same number of examples for cups and cars.
- Try to come up with lots of different types of examples. Make sure that you include some examples with different backgrounds.

Tech flash

- **Machine Learning** : The study of computer algorithms that automatically improve through experience and the use of data. It is a part of artificial intelligence.
- **AI Engines** : An AI engine is a tool that helps build an artificially intelligent system.